



# **Internship Report – Semester Long**

**Activities Undertaken During eSim Internship Autumn 2025**

## **Task 4: eSim Upgradation (CSE and Related Fields)**

**Challenges and Resolutions for Installing eSim on Ubuntu OS 25.04**

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**This report highlights the challenges and issues I faced during the installation of eSim 2.5 on my Ubuntu 25.04 operating system.**

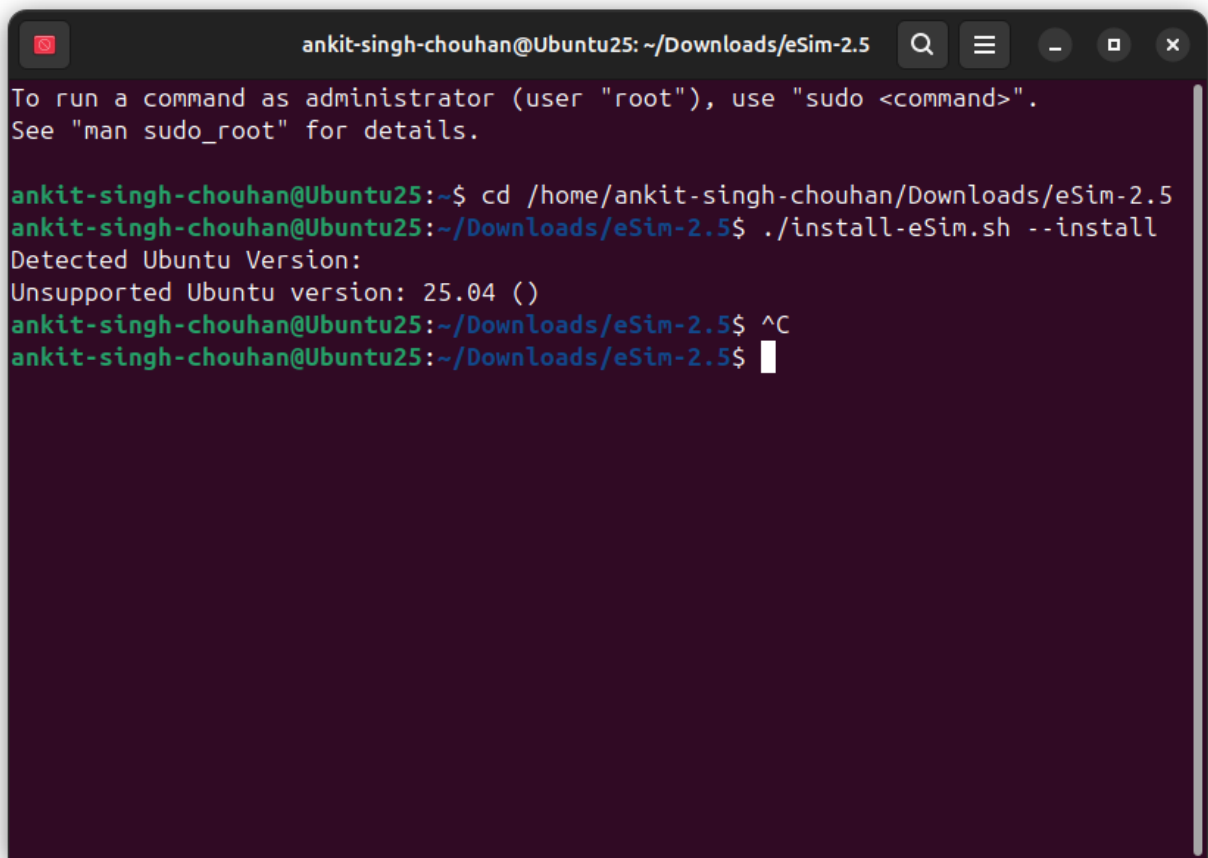
## Problem No. 1:

### Unsupported Ubuntu 25.04.

#### Issue:

While running:

```
./install-eSim.sh --install
```



```
ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ankit-singh-chouhan@Ubuntu25:~$ cd /home/ankit-singh-chouhan/Downloads/eSim-2.5
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ ./install-eSim.sh --install
Detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ ^C
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$
```

The script stopped with:

```
Unsupported Ubuntu version: 25.04 ()
```

## Steps to resolve this issue:

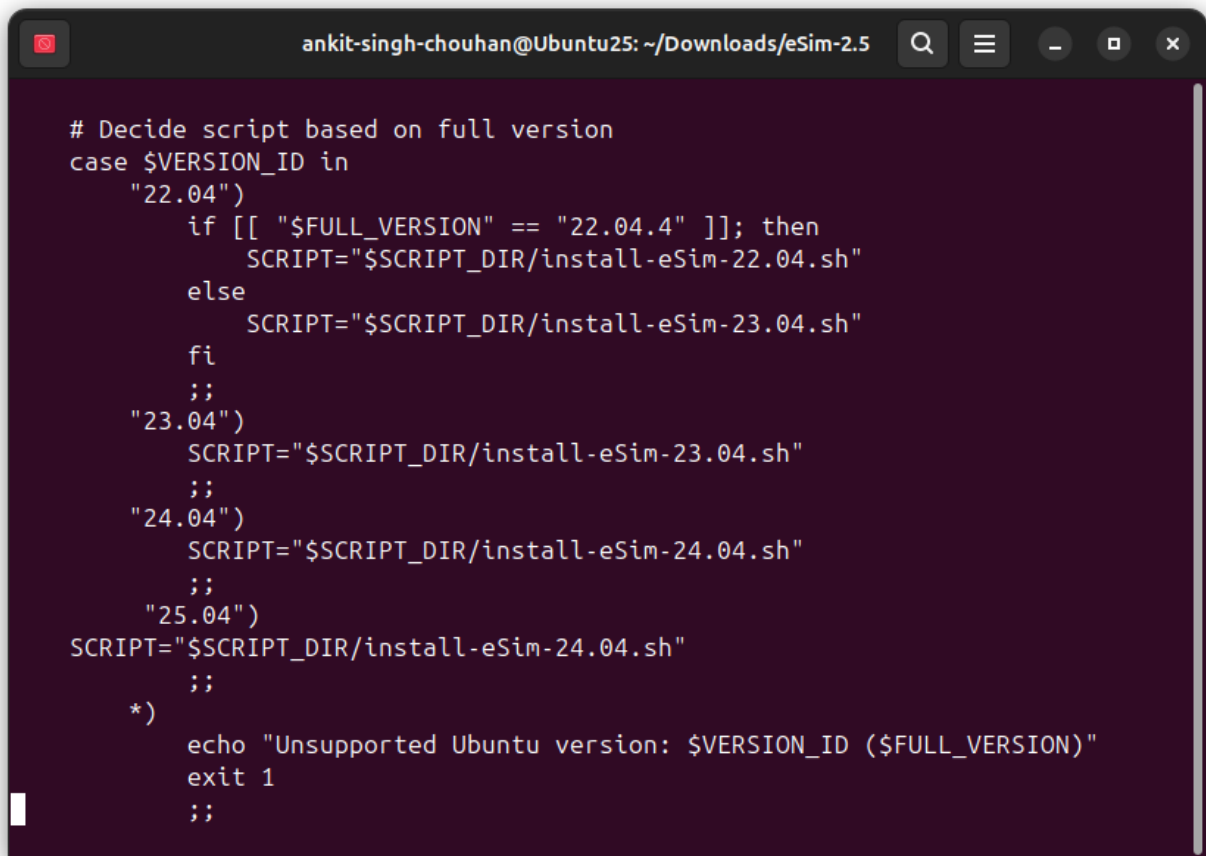
### Step1:

Open the script:

```
vi install-eSim.sh
```

### Step2:

Find the case \$VERSION\_ID in block  
and add this small case for 25.04:

A terminal window with a dark purple background and white text. The window title is 'ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5'. The script content is as follows:

```
# Decide script based on full version
case $VERSION_ID in
    "22.04")
        if [[ "$FULL_VERSION" == "22.04.4" ]]; then
            SCRIPT="$SCRIPT_DIR/install-eSim-22.04.sh"
        else
            SCRIPT="$SCRIPT_DIR/install-eSim-23.04.sh"
        fi
        ;;
    "23.04")
        SCRIPT="$SCRIPT_DIR/install-eSim-23.04.sh"
        ;;
    "24.04")
        SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
        ;;
    "25.04")
        SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
        ;;
    *)
        echo "Unsupported Ubuntu version: $VERSION_ID ($FULL_VERSION)"
        exit 1
        ;;
esac
```

### Step3: Save and exit

## Step4:

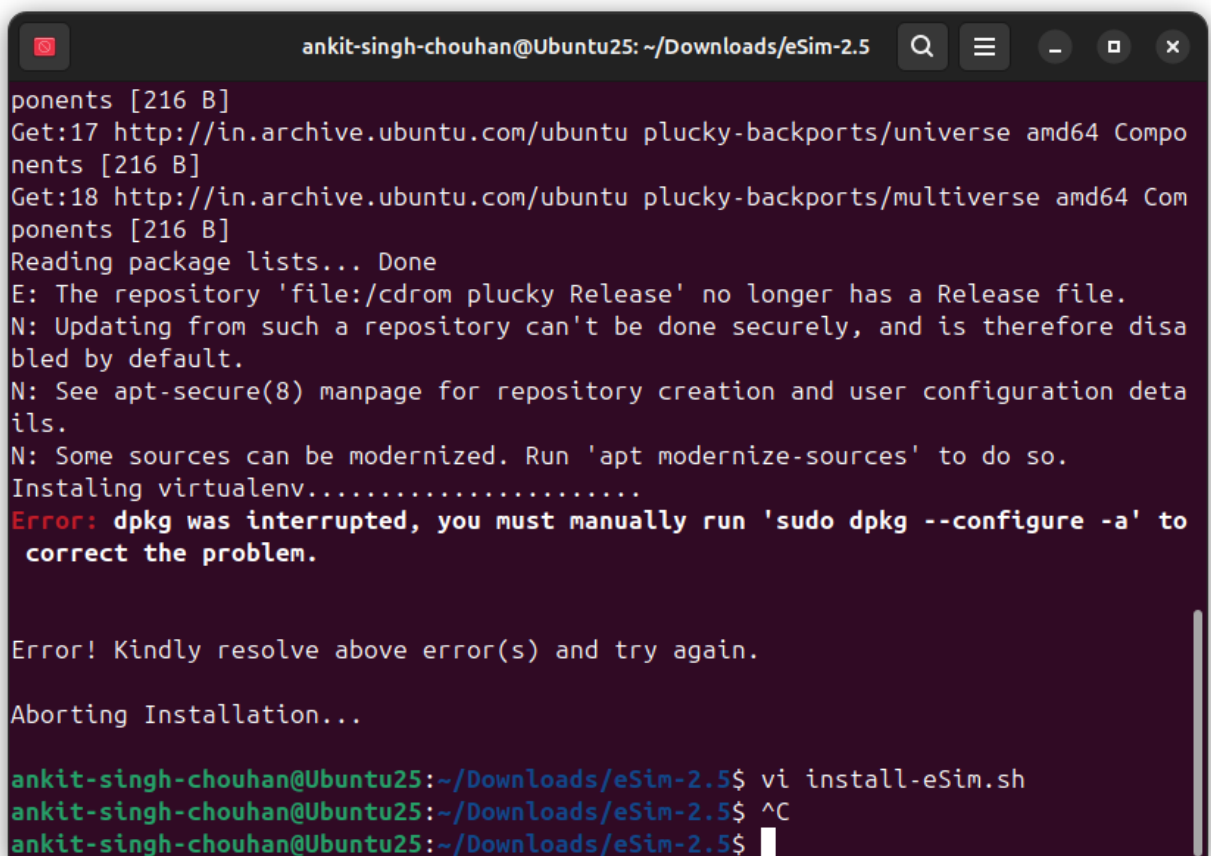
Run installer:

`./install-eSim.sh --install`

## Problem No. 2:

### Repository and dpkg Errors During Installation

**Issue:** While running the eSim installer, the following errors occurred:

A terminal window with a dark background and light text. The window title is 'ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5'. The output shows the installation process for 'virtualenv' failing due to repository errors. The error messages are: 'E: The repository 'file:/cdrom plucky Release' no longer has a Release file.', 'N: Updating from such a repository can't be done securely, and is therefore disabled by default.', 'N: See apt-secure(8) manpage for repository creation and user configuration details.', 'N: Some sources can be modernized. Run 'apt modernize-sources' to do so.', 'Installing virtualenv.....', and a red 'Error: dpkg was interrupted, you must manually run 'sudo dpkg --configure -a' to correct the problem.' followed by 'Error! Kindly resolve above error(s) and try again.' and 'Aborting Installation...'. The prompt then shows the user entering 'vi install-eSim.sh' and pressing '^C' to abort the command.

```
ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5
ponents [216 B]
Get:17 http://in.archive.ubuntu.com/ubuntu plucky-backports/universe amd64 Components [216 B]
Get:18 http://in.archive.ubuntu.com/ubuntu plucky-backports/multiverse amd64 Components [216 B]
Reading package lists... Done
E: The repository 'file:/cdrom plucky Release' no longer has a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
N: Some sources can be modernized. Run 'apt modernize-sources' to do so.
Installing virtualenv.....
Error: dpkg was interrupted, you must manually run 'sudo dpkg --configure -a' to correct the problem.

Error! Kindly resolve above error(s) and try again.

Aborting Installation...

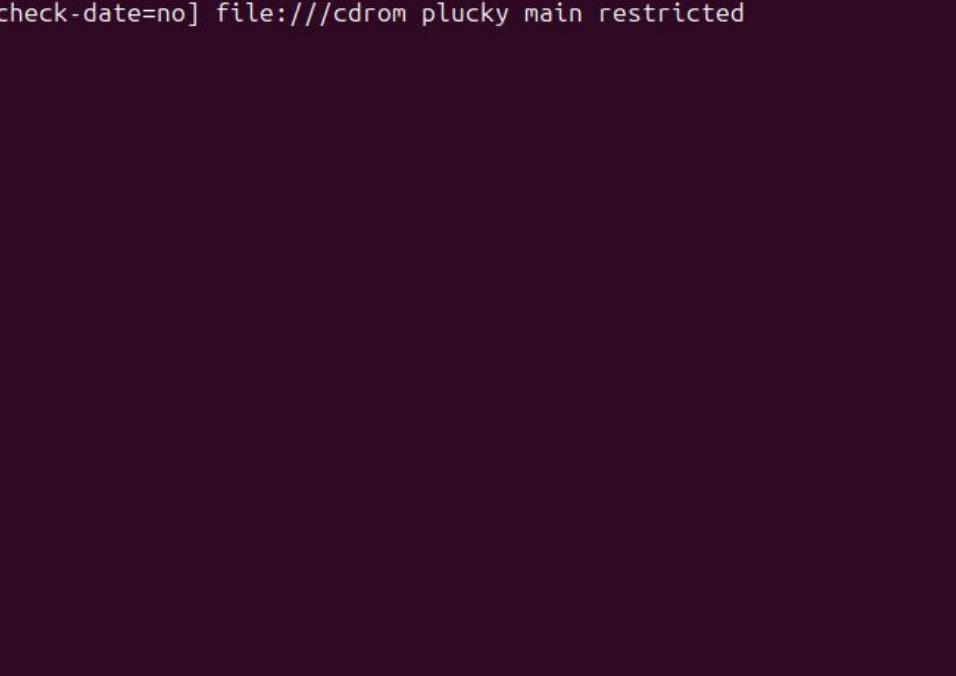
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ vi install-eSim.sh
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ ^C
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$
```

### Steps to resolve this issue:

### Step1:

**Remove the CD-ROM repository:** Open the sources list and comment the cdrom entry:

```
sudo vi /etc/apt/sources.list
```

A terminal window with a dark background and light green text. The window title is "ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5". The terminal shows the command "cat /etc/apt/sources.list" and its output: "deb [check-date=no] file:///cdrom plucky main restricted". At the bottom, it says "1 line, 59 bytes". The terminal has a standard Ubuntu window header with a search icon, a menu icon, and window control buttons (minimize, maximize, close).

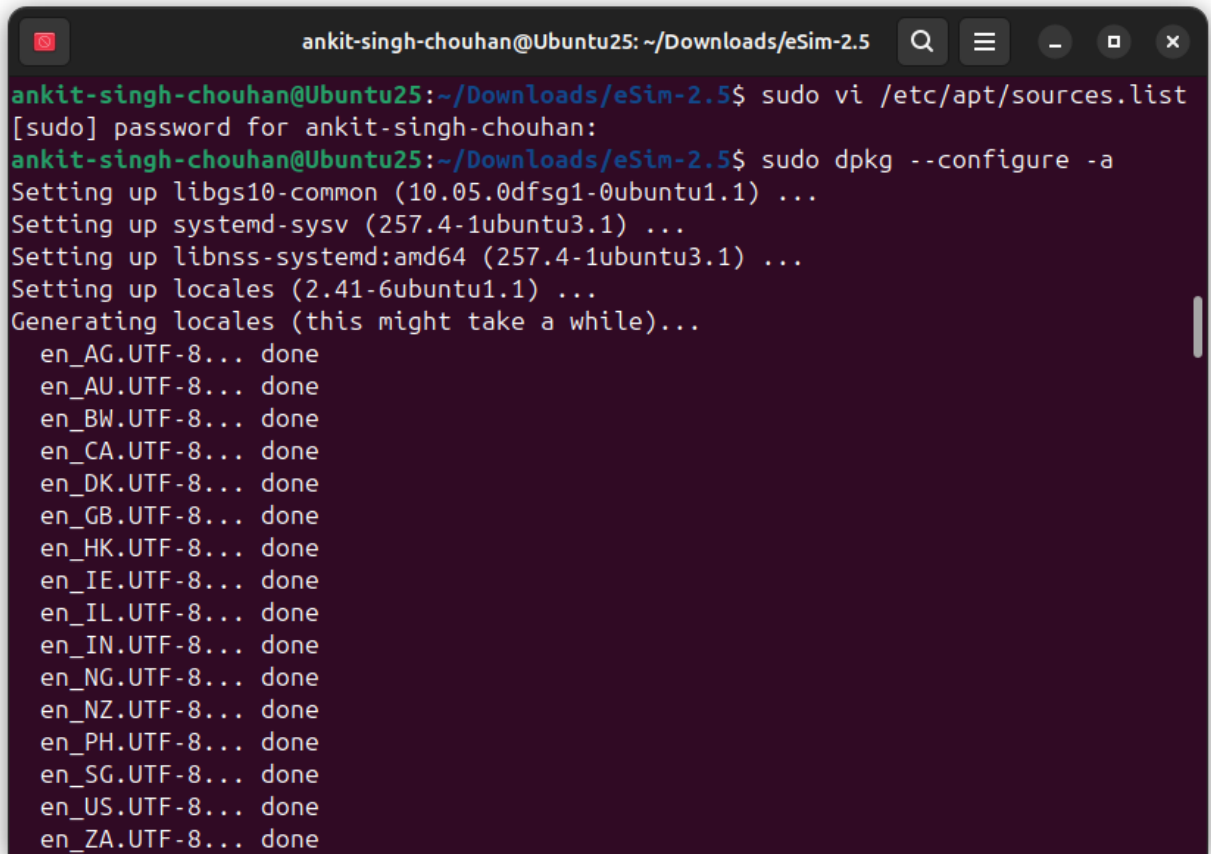
```
ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5
cat /etc/apt/sources.list
deb [check-date=no] file:///cdrom plucky main restricted

1 line, 59 bytes
```

### Step2:

## Reconfigure dpkg:

```
sudo dpkg --configure -a
```

A terminal window with a dark background and light text. The window title is 'ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5'. The user has entered 'sudo vi /etc/apt/sources.list' and 'sudo dpkg --configure -a'. The output shows the configuration of various system packages and the generation of locales. The locales list includes en\_AG.UTF-8 through en\_ZA.UTF-8, all marked as 'done'.

```
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ sudo vi /etc/apt/sources.list
[sudo] password for ankit-singh-chouhan:
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ sudo dpkg --configure -a
Setting up libgs10-common (10.05.0dfsg1-0ubuntu1.1) ...
Setting up systemd-sysv (257.4-1ubuntu3.1) ...
Setting up libnss-systemd:amd64 (257.4-1ubuntu3.1) ...
Setting up locales (2.41-6ubuntu1.1) ...
Generating locales (this might take a while)...
 en_AG.UTF-8... done
 en_AU.UTF-8... done
 en_BW.UTF-8... done
 en_CA.UTF-8... done
 en_DK.UTF-8... done
 en_GB.UTF-8... done
 en_HK.UTF-8... done
 en_IE.UTF-8... done
 en_IL.UTF-8... done
 en_IN.UTF-8... done
 en_NG.UTF-8... done
 en_NZ.UTF-8... done
 en_PH.UTF-8... done
 en_SG.UTF-8... done
 en_US.UTF-8... done
 en_ZA.UTF-8... done
```

This completes any interrupted installations.

### **Step3:**

**Clean & update repositories:**

```
sudo apt clean
```

```
sudo apt update --fix-missing
```

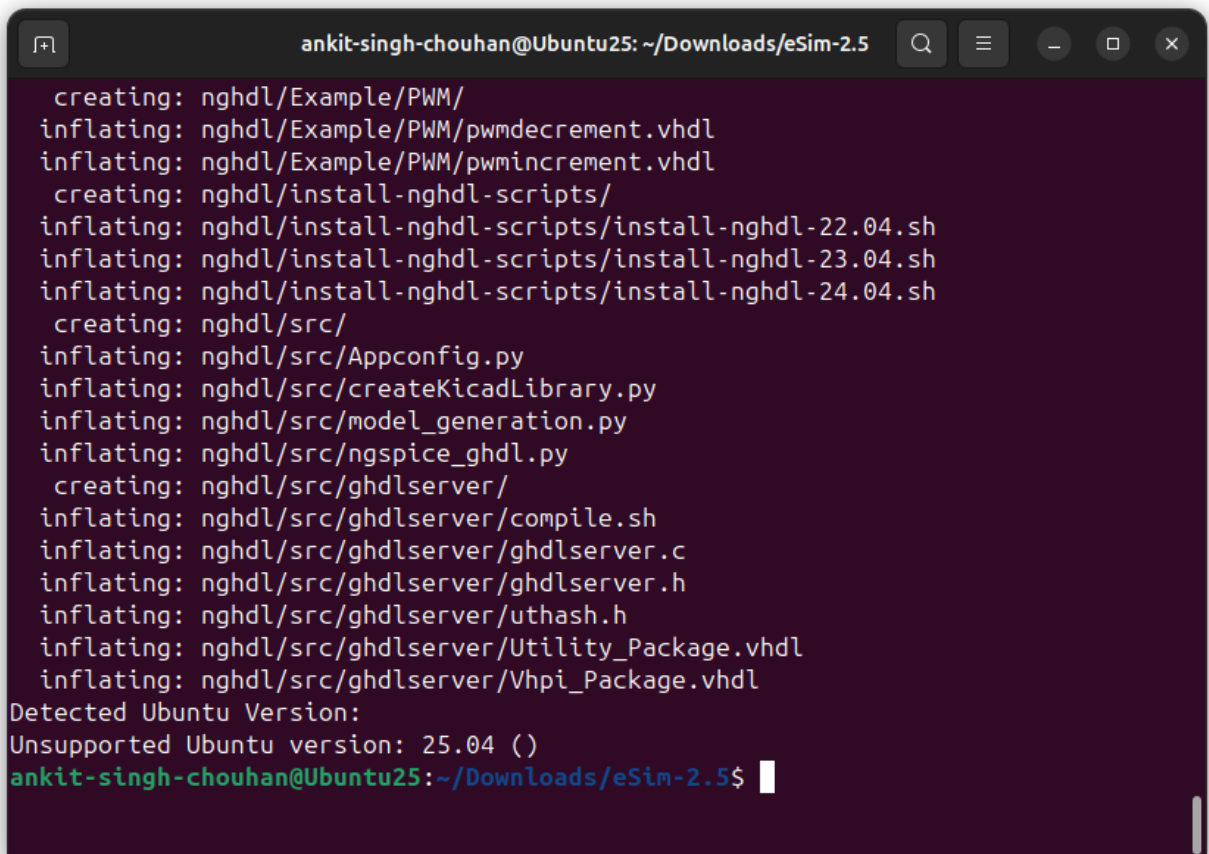
### **Step4:**

**Fix broken packages:**

```
sudo apt --fix-broken install -y
```

### Problem No.3:

**NGHDL installer does not recognize Ubuntu 25.04 and aborts installation.**



```
ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5
creating: nghdl/Example/PWM/
inflating: nghdl/Example/PWM/pwmdecrement.vhdl
inflating: nghdl/Example/PWM/pwmincrement.vhdl
creating: nghdl/install-nghdl-scripts/
inflating: nghdl/install-nghdl-scripts/install-nghdl-22.04.sh
inflating: nghdl/install-nghdl-scripts/install-nghdl-23.04.sh
inflating: nghdl/install-nghdl-scripts/install-nghdl-24.04.sh
creating: nghdl/src/
inflating: nghdl/src/Appconfig.py
inflating: nghdl/src/createKicadLibrary.py
inflating: nghdl/src/model_generation.py
inflating: nghdl/src/ngspice_ghdl.py
creating: nghdl/src/ghdlserver/
inflating: nghdl/src/ghdlserver/compile.sh
inflating: nghdl/src/ghdlserver/ghdlserver.c
inflating: nghdl/src/ghdlserver/ghdlserver.h
inflating: nghdl/src/ghdlserver/uthash.h
inflating: nghdl/src/ghdlserver/Utility_Package.vhdl
inflating: nghdl/src/ghdlserver/Vhpi_Package.vhdl
Detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$
```

**Steps to resolve this issue:**

### **Step1:**

Open the NGHDL install script:

```
nano nghdl/install-nghdl-scripts/install-nghdl.sh
```

### **Step2:**

Locate the Ubuntu version detection code:

### **Step3:**

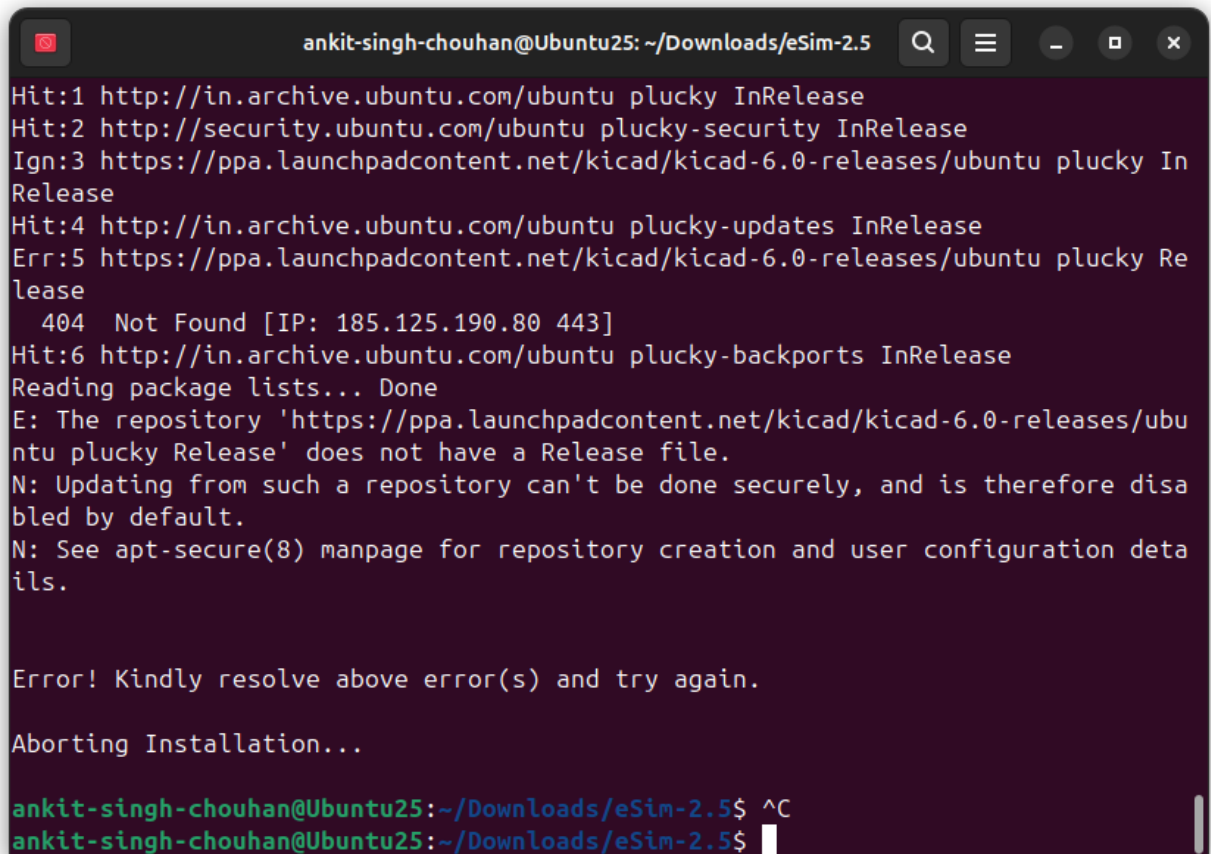
Add 25.04 mapping (use 24.04 script as it's compatible):

```
"25.04") script="install-nghdl-24.04.sh";;
```



## Problem No.4:

### Dependency Conflict While Installing KiCad.

A terminal window titled 'ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5' with standard Ubuntu window controls. The terminal output shows the process of adding repositories and installing KiCad. It lists several 'Hit' messages for repositories like 'plucky', 'plucky-security', 'plucky-updates', and 'plucky-backports'. An 'Ign' message is shown for a KiCad repository. An 'Err' message indicates a 404 Not Found error for the KiCad repository's release file. The terminal then displays error messages from apt, stating that the repository does not have a release file and that updating from such a repository cannot be done securely. The installation is aborted with the message 'Error! Kindly resolve above error(s) and try again.' and 'Aborting Installation...'. The prompt returns to the user, who has pressed the Ctrl-C key.

```
ankit-singh-chouhan@Ubuntu25: ~/Downloads/eSim-2.5
Hit:1 http://in.archive.ubuntu.com/ubuntu plucky InRelease
Hit:2 http://security.ubuntu.com/ubuntu plucky-security InRelease
Ign:3 https://ppa.launchpadcontent.net/kicad/kicad-6.0-releases/ubuntu plucky In
Release
Hit:4 http://in.archive.ubuntu.com/ubuntu plucky-updates InRelease
Err:5 https://ppa.launchpadcontent.net/kicad/kicad-6.0-releases/ubuntu plucky Re
lease
  404 Not Found [IP: 185.125.190.80 443]
Hit:6 http://in.archive.ubuntu.com/ubuntu plucky-backports InRelease
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/kicad/kicad-6.0-releases/ubu
ntu plucky Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disa
bled by default.
N: See apt-secure(8) manpage for repository creation and user configuration deta
ils.

Error! Kindly resolve above error(s) and try again.

Aborting Installation...

ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$ ^C
ankit-singh-chouhan@Ubuntu25:~/Downloads/eSim-2.5$
```